The large-girth theorem with one family of short cycles

A research note on Section 8 of "The maximum number of cycles in a graph of large rank"

Fable 5.2, for KNT — Research note of 12 November 2026

This note answers the research task of 12 November 2026: find a shorter mathematical proof of Theorem 8.1 of the paper [M], the theorem for large girth. The numbers of the statements of [M] are those of the draft of 42 pages. The note gives two new proofs. Each proof deletes one maximal family of short cycles, at one scale. Each proof removes Theorem 8.8 (raising the girth), the nested sets $S_0 \subseteq S_1 \subseteq \cdots$, the two girth scales of Section 8.4, and the deleted set in Propositions 8.5 and 8.7. The manuscript is not changed.

The result in seven sentences

1. In the present proof a block of a chain is a piece only if it avoids a deleted set of short cycles. So the proof needs a small deleted set and a large one, and Theorem 8.8 constructs both.
2. It is sufficient to discard the blocks that *contain* a short cycle. These blocks are disjoint. So their number is at most a packing number: the largest number of pairwise disjoint cycles of length less than $C_0 \log(\Lambda + 2)$.
3. With this change, one proposition on a weighted family of regions gives the packing bound for short cycles and also the final contradiction for cycles of moderate length (Proposition 2.2).
4. The packing bound for the length d then follows from the packing bound for the length $C_0 \log(d+2)$, by induction. Nothing is deleted, and Theorem 8.8 is not necessary (Section 2).
5. A second proof moves this induction into the lemma on pieces. The new lemma says: at a fixed girth D_*, every large connected subcubic multigraph with a small boundary has a forest proportion of less than $\frac{3}{5}$. Then two large disjoint regions of $G - P$ with small boundaries always give $q(G, P) \leq \frac{9}{25}$ (Section 4).
6. Measured in copies of the paper, Section 8 has 271 typeset lines now, 259 lines with the first proof and 289 lines with the second proof. The paper keeps 42 pages. So the gain is in the structure of the proof and not in its length.
7. The hypotheses of the final application do not help (Section 5.3), and each proof keeps an induction over about $\log^* r$ scales (Section 6).

> NOTATION | **the status of this note**
>
> - *Established on paper.* Sections 2 and 4 contain complete proofs. I checked each step against the statements of [M]. Two copies of the paper with the new Section 8 build without an undefined reference.

- *Tested by computer.* Scripts test the facts on joins of Section 4.1 and the examples of Sections 5.3 and 6 on small graphs (Section 7). These tests do not replace a proof.
- *Not done.* There is no formal proof of the new arguments. The Lean proof of [M] follows the present Section 8. No other person or model has read this note.
- *The manuscript.* It is unchanged. Two scripts in the folder of this note apply the first or the second proof to a copy of the paper.

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1 The setting

1.1 Notation and the theorem

Throughout, G is a connected cubic multigraph, ξ is a uniformly random even subgraph of G, and $q(G,P)$ is the probability that $\xi \cap E(G-P)$ is a forest. A *region* is a nonempty set A of vertices such that $G[A]$ is connected. It is *cyclic* if $G[A]$ contains a cycle. The set δA consists of the edges with exactly one end in A; these are the *boundary edges* of A. For disjoint regions A and B, a *link* is an edge between them, or a component of $G-(A\cup B)$ that is joined to both. Their number is $\ell(A,B)$. The event O_A is $\xi \cap \delta A = \emptyset$. Logarithms have base 2, and $\ln$ is the natural logarithm.

Theorem 1.1 ([M, Theorem 8.1]). There are integers $D_* \geq 7$ and r_5 with the following property. Let G be a connected cubic multigraph of rank $r \geq r_5$. Let P be a nonempty set of at most $3\log r$ vertices such that $G[P]$ is connected. If $G-P$ has girth at least D_*, then $q(G,P) < \frac{2}{5}$.

1.2 The present proof, and the place of the two girth scales

The proof of [M] supposes that $q(G,P) \geq \frac{2}{5}$. It has five steps.

1. *Pieces* (Lemmas 8.2 and 8.3). Let $Y' \subseteq V(G)\setminus P$ be a connected set such that $G[Y']$ has at most 2Λ boundary vertices, at least $C_1\Lambda^4$ vertices, and girth at least $C_0\log(\Lambda+2)$. Then $\xi \cap E(G[Y'])$ is a forest with a probability of less than $\frac{3}{5}$, for every choice of the edges outside it. Two disjoint sets of this kind give $q(G,P) \leq \frac{9}{25}$.
2. *A chain* (Lemma 8.4, Proposition 8.5). Pairwise disjoint regions that separate the same two vertices are nested, and the blocks between them have small boundaries. A block must have a large girth. For this purpose the proposition takes a set $U \supseteq P$ such that $G-U$ has large girth, and it discards the blocks that meet $U\setminus P$. So $|U\setminus P|$ must be smaller than the number of blocks.
3. *Packing* (Lemma 8.6, Proposition 8.7). If $G-(P\cup S)$ has girth at least D, then it contains at most $C_2\, d^6(|S|+1)\,2^d$ pairwise disjoint cyclic regions with at most $d \leq 2D$ boundary edges.

4. *Raising the girth* (Theorem 8.8). Step 3 for $D = D_*, 2D_*, 4D_*, \dots$ gives nested sets $S_0 \subseteq S_1 \subseteq \cdots$ such that $G-(P\cup S_j)$ has girth greater than $D_j = 2^j D_*$ and $\log(|S_j|+1) \le 2D_j + C_3 \log^2 D_j$.
5. *Two scales* (Section 8.4). Put $x = \log r$. The second moment of the cycles of moderate length needs the scale $D^+ \approx x/16$, and there $|S^+| = r^{1/4+o(1)}$. The chain of Step 2 has about $r^{1/32}$ members. So its blocks cannot avoid S^+. The proof takes the girth of the blocks from a second scale $D^- \approx \sqrt{x}$, where $|S^-| = r^{o(1)}$.

So the two scales have one cause: Step 2 obtains the girth of a block from a deleted set, and the deleted set of the large scale is larger than the chain. Section 2 removes this cause.

1.3 Statements of [M] that the note uses

Fine print. (a) [M, Lemma 2.4] and its proof. Let $X_1, \dots, X_n$ be pairwise disjoint cyclic regions of a connected multigraph with $|\delta X_i| \le d$. Let E_i be the event that O_{X_i} holds and $\xi \cap E(G[X_i]) \ne \emptyset$, and put $p_i = \mathbb{P}(E_i)$. Then $p_i \ge 2^{-d-1}$, and $\mathbb{P}(E_i \cap E_j) = 2^{\ell(X_i,X_j)-1} p_i p_j$ for $i \ne j$. For each i at most d indices j have $\ell(X_i, X_j) \ge 3$. The probability that no E_i occurs is at most $\frac{1}{2} + (d+1)2^{d-1}/n$.

(b) [M, Section 3.1 and Proposition 3.5]. Let U be a set of vertices such that $J = G - U$ is a simple graph of girth greater than D. Put $s = \lfloor (D-1)/2 \rfloor$. Let $\mathcal{C}$ be the set of the cycles of J of length at most L, with the weights $a_C = 2^{-|C|}$ of total μ. Then, for every vertex w,

$$\sum_{C\in\mathcal{C},\, C\ni w} a_C \;\le\; L\,2^{-s}. \tag{1.1}$$

There is a random variable $Z \ge 0$ with $\mathbb{E}Z = \mu$ such that ξ contains a cycle of J if $Z \ne 0$. Let W_2 be the sum of $a_C a_{C'}$ over the ordered pairs of disjoint cycles of $\mathcal{C}$ whose vertex sets have exactly two links. Proposition 3.5 says the following. Let r be large, $x = \log r$ and $y = \log x$. Let P be a set of at most $2r/x^2$ vertices such that $G[P]$ has at most $x/100$ components, and let $q(G,P) \ge \frac{2}{5}$. Let D and L be integers with $x/16 \le D \le x/8$ and $7x \le L \le x\sqrt{y}$. Let S be the union of the vertex sets of a maximal family of pairwise disjoint cycles of length at most D in $G-P$, and let $|S| \le r/x^2$. Then, for $U = P \cup S$,

$$L^2 2^{-s} \le r^{-1/60}, \qquad \mu \ge \frac{1}{4}\ln\frac{L}{6x} - \frac{1}{\sqrt{x}}, \qquad \mathbb{E}Z^2 \le \mu^2 + W_2 + \bigl(1 + r^{-1/60}\bigr)\mu.$$

(c) [M, Lemma 8.2]. There are absolute constants C_0 and $C_1 \ge 1$ with the following property. Let $\Lambda \ge 4$, and let N be a connected subcubic multigraph with at most 2Λ boundary vertices, at least $C_1\Lambda^4$ vertices, and girth at least $C_0 \log(\Lambda+2)$. Then every set of boundary vertices of even size has forest proportion less than $\frac{3}{5}$. (The boundary vertices are the vertices of degree less than three. For an even set T of them, a T-join is a set of edges in which exactly the vertices of T have odd degree, and the forest proportion of T is the proportion of the T-joins that are forests.)

(d) [M, Lemma 8.3]. Let $U_1, \dots, U_m \subseteq V(G) \setminus P$ be pairwise disjoint sets such that each $G[U_i]$ is connected and every even set of boundary vertices of $G[U_i]$ has forest proportion at most $\frac{3}{5}$. Then $q(G,P) \le (3/5)^m$.

(e) [M, Lemma 8.4]. Let u and p be two vertices of a connected graph H. Pairwise disjoint connected sets that avoid u and p and separate u from p are linearly ordered: $C_1 < C_2$ means that $C_1 \cup U_1 \subseteq U_2$, where U_i is the vertex set of the component of u in $H - C_i$.

(f) [M, Lemma 8.6]. Let p be a vertex of a connected multigraph. Let $\mathcal{A}$ be a finite family of regions that avoid p, with weights $w_A \ge 0$ of total $\mu > 0$. Let W_2 be the sum of $w_A w_{A'}$ over the ordered pairs of disjoint regions of $\mathcal{A}$ with exactly two links. Let N and e be numbers such that, for every $A \in \mathcal{A}$, at most N vertices outside

> A have a neighbour in A, and the regions of $\mathcal{A}$ that are disjoint from A and joined to A by an edge have total weight at most e. Let K be a number with $W_2 \geq \frac{2}{3}\mu^2 - K\mu$ and $\mu > \frac{3}{2}K + 3e$. Then there are a region $A \in \mathcal{A}$ and a vertex $u \notin A$ in the component of p in $G - A$ such that the regions of $\mathcal{A}$ that are disjoint from A, avoid u, and separate u from p in $G - A$ have total weight at least $(\mu/3 - K/2 - e)/N$.

The next lemma is the step "Blocks" in the proof of [M, Proposition 8.5]. Both new proofs use it, so the note states it alone.

Lemma 1.2 (Blocks). Let u and p be two vertices of a connected graph H. Let $C_1 < C_2 < \cdots < C_n$ be pairwise disjoint connected sets that avoid u and p and separate u from p, and let $w \geq 1$ be an integer. Then there are pairwise disjoint sets $Y_l \subseteq V(H)$, one for each integer $l \geq 0$ with $(l+1)w \leq n-1$, such that

(i) Y_l contains $w-1$ of the sets C_i;

(ii) every edge of H that leaves Y_l ends in C_{lw+1} or in $C_{(l+1)w+1}$;

(iii) Y_l lies in the component of u in $H - C_{(l+1)w+1}$.

Proof. Let U_i be the vertex set of the component of u in $H - C_i$. Then $U_i \cup C_i \subseteq U_j$ for $i < j$. For $i < j$ put $X_{ij} = U_j \setminus (U_i \cup C_i)$. An edge of H that leaves U_j ends in C_j, because U_j is a component of $H - C_j$. No edge joins X_{ij} to U_i: its end in X_{ij} would lie in the component U_i of $H - C_i$. So every edge of H that leaves X_{ij} ends in C_i or in C_j. If $i < l < j$, then $C_l \subseteq U_j$, and C_l avoids $U_i \cup C_i \subseteq U_l$; so $C_l \subseteq X_{ij}$. If $i < j \leq i' < j'$, then $X_{ij} \subseteq U_j \subseteq U_{i'}$ is disjoint from $X_{i'j'}$. So the sets $Y_l = X_{lw+1,(l+1)w+1}$ have the three properties. ∎

2 The first proof: count the blocks that contain a short cycle

Let C_0 and C_1 be the constants of [M, Lemma 8.2]. We may suppose that $C_0 \geq 1$, because a larger number C_0 makes the hypothesis of that lemma stronger.

Call a cycle Λ-*short* if its length is less than $C_0 \log(\Lambda + 2)$. Let $\nu_\Lambda(G, P)$ be the largest number of pairwise disjoint Λ-short cycles of $G - P$. Let $D_* \geq 7$ be a fixed integer such that every integer $d \geq D_*$ satisfies

$$\big(C_0 \log(d+2)\big)^{C_0+9} \leq d, \qquad (d+2)^{C_0} \leq 2\, d^{\,C_0}, \qquad d^2 \geq 20\, C_1. \tag{2.1}$$

Then $C_0 \log(d+2) \leq d$. This number D_* replaces the number D_* of [M], which satisfies $D \geq C_0 \log(2D+2)$ for $D \geq D_*$.

Proposition 2.1 (A long chain)**.** Let G be a connected cubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected. Let $p \in P$, let A be a region that avoids P, and let $u \notin A$ be a vertex in the component of p in $G - A$. Let $C_1, \dots, C_T$ be pairwise disjoint regions that avoid P, A and u and separate u from p in $G - A$. Let $\Lambda \geq 4$ be an integer and $m \geq 1$ a number such that A and every C_i have at most Λ boundary edges, and every C_i has at least m vertices. If

$$T \;>\; \big(\Lambda + \nu_\Lambda(G,P) + 2\big)\big(\lceil 2C_1\Lambda^5/m\rceil + 1\big),$$

then $q(G,P) \leq \frac{9}{25}$.

Proof. Each C_i is connected and separates u from p in $G - A$. So it lies in the component H of $G - A$ that contains u and p. By [M, Lemma 8.4] we may assume that $C_1 < C_2 < \cdots < C_T$. Put $w = \lceil 2C_1\Lambda^5/m\rceil + 1$ and $\nu = \nu_\Lambda(G,P)$.

Blocks. Lemma 1.2 gives at least $\Lambda + \nu + 2$ pairwise disjoint sets Y_l. Each of them contains $w - 1$ of the sets C_i, so $|Y_l| \geq (w-1)m \geq 2C_1\Lambda^5$. Every edge of H that leaves Y_l ends in one of two sets C_i, so at most 2Λ edges of H leave Y_l. The set P is connected, it contains p, and it is disjoint from A and from the sets C_i. So P lies in the component of p in $H - C_i$ for every i, and every Y_l avoids P, by (iii).

Discard Y_l if it contains a vertex with a neighbour in A, or if $G[Y_l]$ contains a Λ-short cycle. The set A has at most Λ boundary edges, and the sets Y_l are pairwise disjoint subsets of $V(G) \setminus P$. So this discards at most $\Lambda + \nu$ sets, and two sets remain. Let Y be one of them. The edges of G that leave Y are edges of H, so $|\delta Y| \leq 2\Lambda$, and $G[Y]$ has girth at least $C_0 \log(\Lambda+2)$.

Pieces. Every component of $G[Y]$ has an edge that leaves Y. So there are at most 2Λ components, and one of them has a vertex set Y' with at least $C_1\Lambda^4$ vertices. A boundary vertex of $G[Y']$ is an end of an edge of δY, so there are at most 2Λ of them. By [M, Lemma 8.2], every even set of boundary vertices of $G[Y']$ has forest proportion less than $\frac{3}{5}$. [M, Lemma 8.3], for the two sets Y', gives $q(G,P) \leq \frac{9}{25}$. ∎

The difference from [M, Proposition 8.5] is in the second paragraph of the step "Blocks". There is no set U and no number D, and the conclusion is for the marked set P.

Proposition 2.2 (Heavy families). Let G be a connected cubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected and $q(G,P) \geq \frac{2}{5}$. Let $\mathcal{A}$ be a finite family of regions that avoid P, with weights $w_A \geq 0$ of total μ, and let W_2 be the sum of $w_A w_{A'}$ over the ordered pairs of disjoint regions of $\mathcal{A}$ with exactly two links. Let $\Lambda \geq 4$ be an integer, and let $m \geq 1$, $\kappa > 0$ and $K \geq 0$ be numbers such that

(a) every region of $\mathcal{A}$ has at most Λ boundary edges and at least m vertices;

(b) for every $A \in \mathcal{A}$, the regions of $\mathcal{A}$ that meet A have total weight at most κ;

(c) some random variable $Z \geq 0$ has $\mathbb{E}Z = \mu$ and $\mathbb{E}Z^2 \leq \mu^2 + W_2 + K\mu$, and $Z = 0$ whenever $\xi \cap E(G-P)$ is a forest.

Then

$$\mu \;\leq\; \tfrac{3}{2}K + 3\Lambda\kappa\,\big(\Lambda + \nu_\Lambda(G,P) + 3\big)\big(\lceil 2C_1\Lambda^5/m\rceil + 1\big). \tag{2.2}$$

Proof. Put $\nu = \nu_\Lambda(G,P)$ and $w = \lceil 2C_1\Lambda^5/m\rceil + 1$. We may assume that $\mu > \frac{3}{2}K + 3\Lambda\kappa$. By (c), $\mathbb{P}(Z \neq 0) \leq 1 - q(G,P) \leq \frac{3}{5}$, and the Cauchy–Schwarz inequality $\mu^2 \leq \mathbb{E}Z^2 \cdot \mathbb{P}(Z \neq 0)$ gives $W_2 \geq \frac{2}{3}\mu^2 - K\mu$.

Fix $p \in P$. We apply [M, Lemma 8.6]. Let $A \in \mathcal{A}$. By (a), at most Λ vertices outside A have a neighbour in A, and a region that is joined to A by an edge contains one of them. The regions of $\mathcal{A}$ that contain a given vertex all meet one of them, so their total weight is at most κ by (b). Hence we may take $N = \Lambda$ and $e = \Lambda\kappa$. The lemma gives a region $A \in \mathcal{A}$, a vertex $u \notin A$ in the component of p in $G - A$, and regions of $\mathcal{A}$ of total weight at least $(\mu/3 - K/2 - \Lambda\kappa)/\Lambda$ that are disjoint from A, avoid u, and separate u from p in $G - A$. Take a maximal family $C_1, \ldots, C_T$ of pairwise disjoint regions among them. Each of these regions meets some C_i, so their total weight is at most $T\kappa$ by (b). As $q(G,P) > \frac{9}{25}$, Proposition 2.1 gives $T \leq (\Lambda + \nu + 2)\,w$. So $\mu/3 - K/2 - \Lambda\kappa \leq \Lambda\kappa\,(\Lambda + \nu + 2)\,w$, and the claim follows because $w \geq 1$. ∎

In other words: if a family with the properties (a), (b) and (c) has a total weight above the bound (2.2), then $q(G,P) < \frac{2}{5}$. The proof has two cases. Either the pairs with two links have a small weight, and the second moment gives $q(G,P) < \frac{2}{5}$. Or they give two large disjoint regions with small boundaries and without a short cycle, and the product gives $q(G,P) \leq \frac{9}{25}$.

Proposition 2.3 (Packing). Let G be a connected cubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected. Suppose that $q(G,P) \geq \frac{2}{5}$ and that $G - P$ has girth at least D_*. Let $d \geq D_*$ be an integer. Then

(i) $G - P$ has at most $2^d\, d^{\,C_0+9}$ pairwise disjoint cycles of length at most d;

(ii) $\nu_d(G,P) \leq 2\, d^{\,C_0+1}$.

Proof. We use induction on d. Suppose that (i) holds for every smaller integer that is at least D_*.

The shorter cycles. Let g be the largest integer that is less than $C_0 \log(d+2)$. Then $g < d$ by (2.1), and a cycle is d-short if and only if its length is at most g. If $g < D_*$, then $G - P$ has no such cycle. Otherwise (i) for g and (2.1) give

$$\nu_d(G,P) \;\le\; 2^g g^{\,C_0+9} \;\le\; (d+2)^{C_0}\big(C_0\log(d+2)\big)^{C_0+9} \;\le\; 2\,d^{\,C_0+1}.$$

This is (ii).

The cycles of length at most d. Let $Q_1, \dots, Q_n$ be pairwise disjoint cycles of length at most d in $G - P$, and put $X_i = V(Q_i)$. Every vertex of Q_i lies on exactly one edge outside Q_i, and Q_i has at least D_* vertices. So X_i is a cyclic region with at most d boundary edges and at least D_* vertices. Let E_i be the event that $\mathcal{O}_{X_i}$ holds and $\xi \cap E(G[X_i]) \ne \emptyset$. Put $p_i = \mathbb{P}(E_i)$, let Z be the number of the events E_i that occur, and put $\mu = \mathbb{E}Z$. By Section 1.3(a), $p_i \ge 2^{-d-1}$ and $\mathbb{P}(E_i \cap E_j) = 2^{\ell(X_i,X_j)-1} p_i p_j$ for $i \ne j$, and for each i at most d indices j have $\ell(X_i, X_j) \ge 3$. Let W_2 be the sum of $p_i p_j$ over the ordered pairs (i,j) with $\ell(X_i,X_j) = 2$. Then $\mathbb{E}Z^2 \le \mu^2 + W_2 + (d+1)\mu$: a pair with at most one link gives at most $p_i p_j$, a pair with two links gives $2p_ip_j$, a pair with three or more links gives at most p_i, and the terms with $i = j$ give μ. If $\xi \cap E(G-P)$ is a forest, then no event E_i occurs, and $Z = 0$.

So Proposition 2.2 applies to the sets X_i with the weights p_i, and with $\Lambda = d$, $m = D_*$, $\kappa = 1$ and $K = d+1$; property (b) holds because the sets are pairwise disjoint and $p_i \le 1$. We have $\lceil 2C_1 d^5/D_* \rceil + 1 \le C_1 d^5$, because $D_* \ge 7$, and $d + 3 \le d^{\,C_0+1}$. With (ii),

$$\mu \;\le\; \tfrac{3}{2}(d+1) + 3d\,\big(d+3+2\,d^{\,C_0+1}\big)\,C_1 d^5 \;\le\; 10\,C_1\,d^{\,C_0+7}.$$

Hence $n \le 2^{d+1}\mu \le 20\,C_1\,2^d d^{\,C_0+7} \le 2^d d^{\,C_0+9}$, by (2.1). This is (i). ∎

2.1 Proof of Theorem 1.1

Let G and P be as in the theorem, with the number D_* of (2.1), and suppose that $q(G,P) \ge \frac{2}{5}$. We derive a contradiction when r is large. All limits are for $r \to \infty$, and they are uniform in G and P. Put

$$x = \log r, \qquad A = 6\,e^{124}, \qquad L = \lfloor Ax \rfloor, \qquad D = \lceil x/16 \rceil.$$

Cycles of moderate length. Let S be the union of the vertex sets of a maximal family of pairwise disjoint cycles of length at most D in $G - P$. By Proposition 2.3(i), $|S| \le 2^D D^{\,C_0+10} \le r/x^2$ for large r. We use [M, Proposition 3.5] for G, P, D, L and S. Its hypotheses hold for large r: the set P has at most $3x$ vertices and $G[P]$ is connected, $x/16 \le D \le x/8$, and $7x \le L \le x\sqrt{\log x}$. So we are in the setting of Section 1.3(b), with $U = P \cup S$ and $J = G - U$. With $\eta = r^{-1/60}$ the proposition gives

$$L^2 2^{-s} \le \eta, \qquad \mu \;\ge\; \frac{1}{4}\ln\frac{L}{6x} - \frac{1}{\sqrt{x}} \;>\; 30, \qquad \mathbb{E}Z^2 \;\le\; \mu^2 + W_2 + (1+\eta)\mu.$$

The bound $\mu > 30$ holds because $L/(6x) \ge e^{124} - 1$.

The contradiction. We apply Proposition 2.2 to the vertex sets of the cycles of $\mathcal{C}$, with the weights a_C, and with $\Lambda = L$, $m = x/16$, $\kappa = \eta$ and $K = 1 + \eta$. A cycle $C \in \mathcal{C}$ has at most L vertices, each of them has one edge end outside C, and the girth of J is greater than $D \geq x/16$. So $V(C)$ has at most L boundary edges and at least $x/16$ vertices; this is (a). A cycle of $\mathcal{C}$ that meets C passes through one of the at most L vertices of C, so (b) holds by (1.1). If $\xi \cap E(G - P)$ is a forest, then ξ contains no cycle of J, and $Z = 0$; this is (c). By Proposition 2.3(ii), $\nu_L(G, P) \leq 2L^{C_0+1}$. So

$$\mu \;\leq\; \tfrac{3}{2}(1+\eta) + 3L\eta\left(L + 2L^{C_0+1} + 3\right)\left(\lceil 32C_1L^5/x\rceil + 1\right) \;=\; \tfrac{3}{2} + o(1),$$

because $L \leq Ax$ and $\eta = r^{-1/60}$. This contradicts $\mu > 30$, and Theorem 1.1 is proved. ∎

3 What the first proof replaces

In [M]	In the first proof
Lemmas 8.2, 8.3, 8.4 and 8.6	No change.
Proposition 8.5 (a long chain): a set U such that $G-U$ has girth at least D; it discards the blocks that meet $U \setminus P$; the conclusion is for $q(G, U)$	Proposition 2.1: no set U and no number D; it discards the blocks that contain a short cycle; the conclusion is for $q(G, P)$.
The step "A chain" in the proof of Proposition 8.7, and the paragraphs "Pairs with two links", "The chain" and "The contradiction" of Section 8.4. This is one argument, written two times	Proposition 2.2 (heavy families), used two times.
Proposition 8.7 (packing): a deleted set S, girth at least D outside $P \cup S$, regions with at most $d \leq 2D$ boundary edges, bound $C_2\, d^6(|S| + 1)2^d$	Proposition 2.3: no deleted set, every length $d \geq D_*$, bound $2^d d^{C_0+9}$, proof by induction on d.
Theorem 8.8 (raising the girth): the sets $S_0 \subseteq S_1 \subseteq \cdots$ at the scales $2^j D_*$, with $\log(|S_j| + 1) \leq 2D_j + C_3 \log^2 D_j$	Removed.
Section 8.4: the scales D^- and D^+, the sets $S^- \subseteq S^+$, the estimate (8.1)	One scale $D = \lceil x/16\rceil$ and one set S.
The number D_* with $D \geq C_0 \log(2D + 2)$ for $D \geq D_*$	The number D_* of (2.1). It is larger.

The size. I applied the first proof to a copy of the paper and built the copy. The count is in lines of the text of the PDF file.

	Now	First proof	Second proof (Section 4)
Section 8, typeset lines	271	259	289
The paper, typeset lines	1624	1615	1644
The paper, pages	42	42	42
Numbered statements in Section 8	8	8	9

So the first proof is shorter by 12 lines in Section 8. The paper as a whole is shorter by 9 lines, because Section 1 must say that the formal proof raises the girth in stages. This is not a large

change of the length. The change is in the structure: the deleted sets, the two scales and Theorem 8.8 are no longer in the proof, and the chain argument is written one time.

Remarks.

- The new packing bound is better. Theorem 8.8 gives about $2^{2D} D^{C_3 \log D}$ vertices in the deleted set of the scale D. Proposition 2.3 gives $2^D D^{C_0+10}$.
- The number D_* is larger than before, because (2.1) contains the constant C_1. It is an absolute constant, and [M, Theorem 7.1] holds for every fixed bound of the girth. So the proof of the main theorem of [M] does not change.
- The formal proof follows the present Section 8. If the manuscript takes a new proof, Section 1.4 of [M] must say that the formal proof of Theorem 8.1 uses another method. The table in the two copies has this sentence.
- Proposition 2.1 is used one time, in Proposition 2.2, and [M, Lemma 8.6] is used one time. I did not merge them, because this does not make the proof shorter.

4 The second proof: pieces at a fixed girth

The first proof has an induction, because [M, Lemma 8.2] needs a girth of about $C_0 \log \Lambda$ for 2Λ boundary vertices. This section proves the lemma at a fixed girth. The induction is then in the proof of this lemma, and the other statements have no packing number.

4.1 Joins

Let N be a connected subcubic multigraph. Its *boundary vertices* are the vertices of degree less than three. Let T be an even set of boundary vertices. A *T-join* is a set of edges in which exactly the vertices of T have odd degree. There are $2^{\beta(N)}$ of them, and the *forest proportion* of T is the proportion of the T-joins that are forests. A region A of N is *T-free* if it contains no vertex of T and every component of $N - A$ contains an even number of vertices of T.

Lemma 4.1 (Free regions). Let J be a uniformly random T-join of N. For a region A let E_A be the event that $J \cap \delta A = \emptyset$ and $J \cap E(N[A]) \neq \emptyset$.

(i) Let A be a region without a vertex of T. Some T-join avoids δA if and only if A is T-free.

(ii) If A is T-free, then E_A has the same probability as for a uniformly random even subgraph of N.

(iii) If A and B are disjoint T-free regions, then $E_A \cap E_B$ is empty, or it has the same probability as for a uniformly random even subgraph of N.

So the bound of Section 1.3(a) holds for J and pairwise disjoint T-free cyclic regions: if there are n of them, each with at most d boundary edges, then the forest proportion of T is at most $\frac{1}{2} + (d+1)2^{d-1}/n$.

Proof. (i) A T-join that avoids δA has an even number of vertices of odd degree in each component of $N - A$. In the other direction, each component K of $N - A$ is connected and has a $(T \cap K)$-join if $|T \cap K|$ is even. The union of these joins is a T-join F without an edge that has an end in A.

(ii) The map $J \mapsto J + F$ is a bijection from the T-joins to the even subgraphs of N, and it keeps the edges that have an end in A. The event E_A depends on these edges only.

(iii) Suppose that some T-join is in $E_A \cap E_B$. Delete its edges in $N[A]$ and in $N[B]$. These form even subgraphs, so the result is a T-join F' without an edge that has an end in $A \cup B$. Use the map $J \mapsto J + F'$.

For the last statement, the proof of [M, Lemma 2.4] uses the probabilities of the events E_i and upper bounds for the probabilities of the events $E_i \cap E_j$. By (ii) and (iii) these hold for J. If J is a forest, then no event E_i occurs. ■

Lemma 4.2 (Product, for joins). Let N be a multigraph, and let P and T be sets of vertices such that N has a T-join. Let $U_1, \dots, U_m \subseteq V(N) \setminus (P \cup T)$ be pairwise disjoint sets of vertices of degree three such that each $N[U_i]$ is connected. Suppose that in each $N[U_i]$ every even set of boundary vertices has forest proportion at most $\frac{3}{5}$. Then $J \cap E(N - P)$ is a forest for at most the proportion $(3/5)^m$ of the T-joins J of N.

For a cubic multigraph G and $T = \emptyset$ this is [M, Lemma 8.3]. The proof is the proof of that lemma.

Proof. Let E' consist of the edges that lie in none of the graphs $N[U_i]$, and let J be a uniformly random T-join. Fix a set $x \subseteq E'$ that extends to a T-join. For each i, let T_i be the set of the vertices of U_i that are incident with an odd number of edges of x. The set U_i avoids T, so the parity equations say that $J \cap E(N[U_i])$ is a T_i-join. These equations involve disjoint sets of edges, so the joins are independent and uniform, conditional on $J \cap E' = x$. A vertex of degree three in $N[U_i]$ has no incident edge in E'. So T_i is an even set of boundary vertices. If $J \cap E(N - P)$ is a forest, its restriction to every $N[U_i]$ is a forest. The conditional probability is therefore at most $(3/5)^m$. Average over x. ■

4.2 The lemma on pieces

Let B be the absolute constant of [M, Section 8.1]: each layer in the conclusion of [M, Lemma 4.1] with $\delta = \frac{1}{20}$, together with at most k marked vertices, has fewer than $B(k+1)^2$ vertices. For $k \geq 1$ let s_k be the least integer with $2^{s_k} > (2s_k+1)B(k+1)^2$. Then $2^{s_k} \leq 2(2s_k+1)B(k+1)^2$, the numbers s_k do not decrease, and $s_k = O(\log k)$.

Lemma 4.3 (Pieces with few short cycles)**.** Let $k \geq 2$, and let N be a connected subcubic multigraph with at least two and at most k boundary vertices. Put $s = s_k$. Let $\mathcal{D}$ be a maximal family of pairwise disjoint cycles of length at most $2s$ in N, and let t be their number. If N has more than $3 \cdot 2^s(k + 2s\,t)$ vertices, then every set of boundary vertices of even size has forest proportion less than $\frac{3}{5}$.

The lemma has no hypothesis on the girth. For $t = 0$ it is [M, Lemma 8.2]: there $k = 2\Lambda$, the girth is greater than $2s$, and $C_1\Lambda^4 > 6\Lambda\, 2^s$.

Proof. Fix a boundary vertex o, and call the set of the vertices at distance j from o the *layer* j.

A large layer. A ball of radius s has at most $3 \cdot 2^s$ vertices. So some vertex w has distance more than s from every boundary vertex and from every vertex of a cycle of $\mathcal{D}$. Let W be the set of the vertices within distance s of w. They have degree three. Take a breadth-first tree from w. Suppose that some vertex at distance less than s from w has fewer than two children in this tree, or that w has fewer than three. Then $N[W]$ contains a cycle of length at most $2s$. By the maximality of $\mathcal{D}$, this cycle meets a cycle of $\mathcal{D}$, and this contradicts the choice of w. So $|W| \geq 2^s$. The distances from o to two vertices of W differ by at most $2s$. So W meets at most $2s + 1$ layers, and some layer has at least $2^s/(2s + 1) > B(k + 1)^2$ vertices.

A marker. This step is the second step of the proof of [M, Lemma 8.2]. Suppose that some set T of boundary vertices has forest proportion at least $\frac{3}{5}$. Take a breadth-first tree of N from o, and let P be the union of its paths from o to the other boundary vertices. Then $N[P]$ is connected, every vertex outside P has degree three, and P has at most k vertices in each layer. The tree paths inside P contain a T-join F_0. The map $J \mapsto J + F_0$ is a bijection from the T-joins of N to the even subgraphs of N, and $J + F_0$ has the same edges in $N - P$ as J. So if J is a forest, then $J + F_0$ is a forest in $N - P$. Hence $q(N, P) \geq \frac{3}{5} > \frac{1}{2} + \frac{1}{20}$. The distance from o is a level function. By [M, Lemma 4.1], each layer has fewer than $B(k + 1)^2$ vertices. This contradicts the first step. ∎

Theorem 4.4 (Pieces at a fixed girth)**.** There are an integer $D_* \geq 7$ and an absolute constant $C_1 \geq 2$ with the following property. Let $k \geq 1$, and let N be a connected subcubic multigraph of girth at least D_* with at most k boundary vertices and at least C_1k^7 vertices. Then every set of boundary vertices of even size has forest proportion less than $\frac{3}{5}$.

Proof. Fix k_0 such that $4s_k < k$ for all $k \geq k_0$, and put $D_* = 2s_{k_0} + 1$. We define numbers $F(k) \geq 2$ below, and we prove the theorem with $F(k)$ in the place of C_1k^7, by induction on k.

Let N and k be as in the theorem, with at least $F(k)$ vertices, and put $s = s_k$. If N has at most one boundary vertex, the only set is the empty one. Then N contains a cycle, because it has at least two vertices, and the forest proportion is at most $\frac{1}{2}$. So let N have at least two boundary vertices, and let $\mathcal{D}$ and t be as in Lemma 4.3.

The base. Let $k < k_0$. Then $2s < D_*$, so $t = 0$. By Lemma 4.3 the theorem holds with $F(k) = 3 \cdot 2^s k + 1$.

The numbers. Let $k \geq k_0$, and put $k' = 4s$. Then $k' < k$. Put

$$w = \lceil k'F(k')/D_* \rceil + 1, \qquad n_1 = 5(2s+1)\,2^{2s}, \qquad n_2 = k^2(k+2)\,w,$$
$$F(k) = 3 \cdot 2^s \big(k + 2s\,(k + n_1 + n_2 + 2)\big) + 1.$$

If $t < k + n_1 + n_2 + 2$, the claim holds by Lemma 4.3. So let $t \geq k + n_1 + n_2 + 2$, and let T be an even set of boundary vertices. At most k cycles of $\mathcal{D}$ contain a boundary vertex. Each of the other cycles Q has at most $2s$ boundary edges, and $V(Q)$ contains no vertex of T. Call Q *free* if $V(Q)$ is T-free. So more than n_1 cycles are free, or more than n_2 cycles are not free.

More than n_1 free cycles. By Lemma 4.1 with $d = 2s$, the forest proportion of T is less than $\frac{1}{2} + (2s+1)\,2^{2s-1}/n_1 = \frac{3}{5}$.

More than n_2 cycles that are not free. Let Q be one of them. Some component of $N - V(Q)$ contains an odd number of vertices of T. The set T is even and avoids Q. So T has a vertex a in this component and a vertex b in another component, and $V(Q)$ separates a from b. There are fewer than k^2 pairs a, b. So more than $(k+2)w$ of these cycles separate the same two vertices a and b. By [M, Lemma 8.4] their vertex sets are linearly ordered, and Lemma 1.2 for N, a and b gives $k+2$ pairwise disjoint sets Y_l. At most k of them contain a boundary vertex of N. Let Y_1 and Y_2 be two of the others, and let $i \in \{1, 2\}$. The set Y_i contains $w - 1$ of the cycles, and each cycle has at least D_* vertices. So $|Y_i| \geq (w-1)D_* \geq k'F(k')$. At most $2 \cdot 2s = k'$ edges leave Y_i, and every component of $N[Y_i]$ has such an edge, because $b \notin Y_i$. So some component has a vertex set Y_i' with at least $F(k')$ vertices. The vertices of Y_i have degree three in N, so $N[Y_i']$ has at most k' boundary vertices. By induction, every even set of boundary vertices of $N[Y_i']$ has forest proportion less than $\frac{3}{5}$. Lemma 4.2, for N, T, $P = \emptyset$ and the disjoint sets Y_1' and Y_2', shows that the forest proportion of T is at most $\frac{9}{25}$.

The size of F. We have $2^s = O(k^2 \log k)$ and $s = O(\log k)$. So $n_1 = O(k^4 \log^3 k)$ and $n_2 = O\big(k^3 \log k \cdot F(4s_k)\big)$, and the definition gives $F(k) = O\big(k^6 \log^5 k + k^5 \log^3 k \cdot F(4s_k)\big)$. Hence $F(k) \leq C_1 k^7$ for an absolute constant C_1, by induction on k. ∎

Corollary 4.5 (Two regions). Let G be a cubic multigraph, and let P be a set of vertices such that $G - P$ has girth at least D_*. Suppose that $V(G) \setminus P$ contains two disjoint sets Y_1, Y_2 and that, for some $k_1, k_2 \geq 1$, each $G[Y_i]$ is connected, $|\delta Y_i| \leq k_i$ and $|Y_i| \geq C_1 k_i^7$. Then $q(G, P) \leq \frac{9}{25}$.

Proof. The multigraph $G[Y_i]$ has at most k_i boundary vertices and girth at least D_*. Use Theorem 4.4 and Lemma 4.2 with $T = \emptyset$. ∎

4.3 Section 8 in the second form

Let D_* and C_1 be as in Theorem 4.4. The statements of Section 2 change as follows. In each of them, $G - P$ has girth at least D_*.

- *A long chain.* For an integer $\Lambda \geq 1$: if $T > (\Lambda+2)\big(\lceil 2^8 C_1 \Lambda^8/m\rceil + 1\big)$, then $q(G,P) \leq \frac{9}{25}$. In the proof, put $w = \lceil 2^8 C_1 \Lambda^8/m\rceil + 1$. The only sets Y_l to discard are the at most Λ sets that contain a vertex with a neighbour in A. A set Y that remains has $|\delta Y| \leq 2\Lambda$ and $|Y| \geq (w-1)m \geq 2\Lambda \cdot C_1(2\Lambda)^7$. So a component of $G[Y]$ has at least $C_1(2\Lambda)^7$ vertices, and Corollary 4.5 applies with $k_1 = k_2 = 2\Lambda$.
- *Heavy families.* The bound is $\mu \leq \frac{3}{2}K + 3\Lambda\kappa\,(\Lambda+3)\big(\lceil 2^8 C_1 \Lambda^8/m\rceil + 1\big)$. The proof does not change.
- *Packing.* For every integer $d \geq D_*$, the multigraph $G - P$ has at most $2^{10} C_1\, 2^d d^{10}$ pairwise disjoint cycles of length at most d. The proof is the second step of the proof of Proposition 2.3, and there is no induction: with $\Lambda = d$, $m = D_*$, $\kappa = 1$ and $K = d+1$,

$$\mu \;\leq\; \tfrac{3}{2}(d+1) + 3d\,(d+3)\big(\lceil 2^8 C_1 d^8/D_*\rceil + 1\big) \;\leq\; 2^9 C_1\, d^{10}, \qquad n \leq 2^{d+1}\mu.$$

- *The proof of the theorem.* Now $|S| \leq 2^{10} C_1\, 2^D D^{11} \leq r/x^2$, and the last display is $\mu \leq \frac{3}{2}(1+\eta) + 3L\eta\,(L+3)\big(\lceil 2^{12} C_1 L^8/x\rceil + 1\big) = \frac{3}{2} + o(1)$.

In this form the numbers ν_Λ, the condition (2.1), and part (ii) of Proposition 2.3 do not exist. The price is the longer proof of Theorem 4.4, and the section is longer by 18 lines than it is now (Section 3).

5 The questions of the task

5.1 Priority 1: one family of short cycles at one scale

(1) Is there a direct bound for the number of vertices that we remove? Yes. Proposition 2.3 gives $|S| \leq 2^D D^{\,C_0+10}$ from $q(G,P) \geq \frac{2}{5}$ and the girth D_*, for one maximal family at the scale D. In the first proof, "direct" has one restriction: the bound for the length d uses the bound for the length $C_0 \log(d+2)$. In the second proof the packing bound has no induction.

(2) Do the pairs of removed short cycles with two links give the long chain? Yes. This is the proof of Proposition 2.3: the family of the short cycles is a family for Proposition 2.2. The proof of [M, Proposition 8.7] already has this idea. The new point is that no deleted set is necessary for the girth of the blocks.

(3) Can we use the separators before the logarithmic girth is known? Yes. Proposition 2.2 has no hypothesis on the girth of the family. The girth is in two places only: in the number $\nu_\Lambda(G,P)$ of the blocks that we discard, and in [M, Proposition 3.5], which controls the second moment of the cycles of moderate length.

(4) Where does the present argument need two scales, and is this necessary? Section 1.2 gives the place: Proposition 8.5 takes the girth of a block from a deleted set, and the deleted set of the scale $x/16$ has more vertices than the chain has members. It is not necessary. A block needs no short cycle *inside* it, and the number of the blocks with a short cycle is at most $\nu_L(G,P) = (\log r)^{O(1)}$.

5.2 Priority 2: one structural lemma

The task asks for a lemma of this form: either the forest probability is below $\frac{2}{5}$, or the graph contains two large disjoint regions with small boundaries. Proposition 2.2 with Corollary 4.5

is this lemma. Suppose that a family has the properties (a), (b) and (c) and that its weight is above the bound. Then

- the pairs with two links have weight $W_2 < \frac{2}{3}\mu^2 - K\mu$, and $q(G,P) < \frac{2}{5}$ by the second moment; or
- $V(G) \setminus P$ contains two disjoint connected sets with at most 2Λ boundary edges and at least $C_1(2\Lambda)^7$ vertices, and $q(G,P) \le \frac{9}{25}$ by Corollary 4.5.

The regions come from the pairs with two links, through [M, Lemma 8.6] and Lemma 1.2. No packing bound is used.

The parts of this lemma are not all new. The second moment, [M, Lemma 8.6] and the blocks are in [M]. Three points are new: the statement for a general weighted family, which serves the short cycles and the cycles of moderate length; the count of the blocks with a short cycle in the place of a deleted set (first proof); and Theorem 4.4, which removes the condition on the girth of the regions (second proof).

5.3 Priority 3: the hypotheses of the final application

In Case 5 of [M, Section 9], G has no cyclic cut of size at most four, its girth is at least D_*, and P is a connected set of eight vertices. The task asks if these hypotheses exclude the pairs with two links. They do not.

Lemma 5.1 (Necklaces). Let R be a bipartite cubic graph with $n \ge 10$ vertices and a Hamiltonian cycle $v_0v_1\cdots v_{n-1}$, and let $M \ge 4$. Let G_M be the graph on the vertices (v_a, i) with $i \in \mathbb{Z}_M$ that has the edges $(v_a, i)(v_{a+1}, i)$ and, for each chord v_av_b of the Hamiltonian cycle with a even, the edges $(v_a, i)(v_b, i+1)$. The *ring* Z_i is the cycle on the vertices (v_a, i). Then

(i) G_M is a connected cubic graph, and its girth is at least the girth of R;

(ii) G_M has no cyclic cut of size at most four;

(iii) two rings that are not consecutive have exactly two links.

Proof. (i) The map $(v_a, i) \mapsto v_a$ is a covering map from G_M to R. So a cycle of G_M maps to a closed walk of R without an immediate return, and such a walk has at least as many edges as a shortest cycle of R.

(ii) Each vertex is on one edge outside its ring. For a vertex (v_a, i) with a even this edge goes to Z_{i+1}, and for a odd it goes to Z_{i-1}. So $n/2 \ge 5$ edges join two consecutive rings, and a ring is an induced cycle. Let X be a set of vertices with $|\delta X| \le 4$ such that X and its complement are not empty. We show that $G[X]$ or $G - X$ has no cycle. Call a ring *split* if it has vertices on both sides. A cut contains an even number of edges of a cycle. So δX contains at least two edges of a split ring, and at most two rings are split. Two consecutive rings that are not split are on the same side.

If no ring is split, all rings are on one side. This is not possible. If one ring is split, the other rings are consecutive, so they are on one side. The other side is a proper subset of a ring,

and it induces no cycle. Let two rings be split. Then δX contains two edges of each of them and no edge between two rings. Suppose that the split rings are not consecutive, and let Z_j be one of them. Its vertices with a even are on the side of Z_{j+1}, and its vertices with a odd are on the side of Z_{j-1}. If these sides are equal, Z_j is not split. If they differ, all edges of Z_j are in δX. Both cases are not possible. So the split rings are Z_j and Z_{j+1}. The other rings are consecutive and on one side. Let Y be the other side. A vertex of Z_j with a odd has an edge to Z_{j-1}, and this edge is not in δX. So the vertices of Y in Z_j have a even. They are consecutive on Z_j, because δX contains two edges of Z_j. So there is one of them. In the same way, with the edges to Z_{j+2}, the set Y has one vertex in Z_{j+1}. So $|Y| = 2$, and Y induces no cycle.

(iii) Without two rings that are not consecutive, the graph has two components. Each of them is joined to the two rings, and no edge joins the two rings. ■

The Heawood graph ($n = 14$, girth 6) and the Tutte–Coxeter graph ($n = 30$, girth 8) are bipartite, cubic and Hamiltonian. For every g there is such a graph R of girth at least g. For example, take a cover of R with the group $\mathbb{Z}_k$ in which one edge of the Hamiltonian cycle has the voltage 1 and the chords have the voltages $(2g)^1, (2g)^2, \dots$, for a large k. The cover is bipartite and cubic, and the Hamiltonian cycle lifts to a Hamiltonian cycle. A cycle of the cover with fewer than g edges maps to a closed walk of R whose voltages add up to zero. For a large k this is possible only if the walk uses each of these edges the same number of times in the two directions. A shortest cycle of R is not such a walk, because it uses a chord one time, or it is the Hamiltonian cycle. So the girth increases, and we repeat the step.

So there are cubic graphs of every girth, without a cyclic cut of size at most four, that contain M pairwise disjoint cycles of one length n in which all pairs that are not consecutive have exactly two links. For the family of these M rings, with equal weights, $W_2 = (1 - 3/M)\mu^2$. The second moment of this family gives no bound below $\frac{1}{2}$. For $M > 2^{16n}$ the rings are cycles of length less than $(\log r)/16$, so they are short cycles at the scale of the proof. The theorem holds for these graphs. But a proof must find the two pieces between the rings, and the hypotheses do not remove this case. The reason is that the hypothesis is on cuts of size at most four, and the regions of the proof have at least $D_* \geq 7$ boundary edges.

Two more remarks on the final application.

- Case 5 needs less than $q(G, P) < \frac{2}{5}$. For a connected set P of p vertices the argument of [M, Section 9] gives $c(G) \leq \big(q(G, P) + 2^{-1-p/2}\big)2^r$. So a bound $q(G, P) \leq \frac{1}{2} - \varepsilon$, for one fixed $\varepsilon > 0$ and connected sets of a fixed size, is sufficient. I found no simpler proof for a weaker constant: one piece gives no bound below $\frac{1}{2}$, and two pieces give $\frac{9}{25}$.
- The proofs of this note use the set P in two ways only: it contains the vertex p, and the regions avoid it. The size of P is used in [M, Proposition 3.5] only.

6 What stays, and why

The packing bound stays. Proposition 8.7 of [M] is not removed; it has a shorter statement and no deleted set. It cannot be removed in this method. The second moment of [M, Proposition 3.5]

needs a graph J of girth about $(\log r)/16$: the number of the links of two cycles of length about $\log r$ is controlled by the girth (Step 2 of that proof). And the weight of the cycles of J is large only if few edges join J to the deleted set ([M, Lemma 3.4] needs $L\, d_J/v = o(1)$). So the case of many disjoint short cycles is a separate case, and the packing bound is this case.

An induction over scales stays. There are three facts.

1. The second moment alone gives the bound $\frac{1}{2}$ and no more, because two regions with two links have a correlation of two. A bound below $\frac{1}{2}$ needs two pieces.
2. A piece needs a large layer. In the proof of [M, Lemma 8.2] the layer comes from a ball that is a tree, of radius about $2\log k$ for k boundary vertices. So a block with a boundary of size k must be free of cycles of length about $4\log k$, or we must count such blocks.
3. [M, Theorem 7.1] treats the graphs with a cycle shorter than a fixed number. Its error terms come from the forest estimate with $\varepsilon = 2^{-g-2}$. So it cannot treat a girth that grows with r.

The girth of the hypothesis is the constant D_*, and the final step needs blocks without cycles of length about $C_0 \log\log r$. An argument must pass from the one to the other. In [M] it is the sequence of the scales $2^j D_*$, with about $\log\log r$ steps and nested deleted sets. In the first proof it is the induction $d \mapsto C_0 \log(d+2)$ in Proposition 2.3, with about $\log^* r$ steps and no deleted set. In the second proof it is the induction $k \mapsto 4s_k$ in Theorem 4.4. I found no proof without such an induction.

The factor two is a parity. The script `checks/necklace_parity.py` examines a necklace of M rings of length four, for $4 \le M \le 7$, by complete enumeration. Let the *parity* of ξ be the parity of its number of edges between two fixed consecutive rings. The parity is even with probability $\frac{1}{2}$. If it is odd, no ring is a component of ξ. If it is even, each ring is a component with two times its probability without the condition, and for rings that are pairwise not consecutive the probability that none of them is a component is the product, as for independent events. This is a test on one family of graphs and not a theorem. It shows why the second moment stops at $\frac{1}{2}$, and why the blocks of Theorem 4.4 need another argument when the parity is odd: there the cycles of the chain are not free.

The lemma on pieces is false for girth four. In a ladder with n rungs, let T consist of the two ends of one rail. Then every T-join is a path, for every n. So the forest proportion is 1, and the ladder has four boundary vertices and any number of vertices (`checks/tjoin_check.py` tests $n \le 10$). In the proof of Theorem 4.4 the ladder is the case of cycles that are not free, and the induction does not start: the cycles of length four have four boundary edges, and the pieces between them have four boundary vertices again. The proof of Theorem 4.4 needs $D_* = 2s_{k_0}+1$, which is more than 200.

An open question, and what it would remove. I do not know the least girth for which Theorem 4.4 holds. In its proof the girth is used in the base only, for pieces with fewer than k_0 boundary vertices. So the theorem holds at a smaller girth if it holds there for these pieces, with some bound for the number of vertices. The pieces of the application are induced subgraphs of a cubic multigraph without a cyclic cut of size at most four, and such a multigraph has girth at least five. Suppose that the base holds for these pieces at girth five. Then Theorem 1.1 holds

with $D_* = 5$ in the final application, Case 4 of [M, Section 9] is empty, and [M, Section 7] with [M, Lemma 5.5] and [M, Corollary 5.6] is not necessary. For two boundary vertices, [M, Corollary 5.6] is a statement of this kind, with $\frac{1}{2} + \varepsilon$ in the place of $\frac{3}{5}$. For more boundary vertices I have no proof and no counterexample. This is a question and not a result.

Approaches that failed.

- *A marker without layers.* If a small connected set contains all boundary vertices of a block, then [M, Theorem 3.1] gives a forest proportion of at most $\frac{1}{2} + o(1)$ at once, with no hypothesis on the girth. For a block between two cycles C_i and C_j the set is the two cycles and a path between them. The obstruction is a long thin block: the path has almost as many vertices as the block. The ladder is the extreme case.
- *A second moment in a block.* The cycles of the chain that lie in one block are pairwise disjoint, so their total weight is at most η. They give no bound.
- *A chain that is longer than a trivial count.* The proof gives a chain of about $2^s/L^3$ members, and this is less than $r^{1/16}$ for a girth of at most $x/8$. Without a packing bound, the number of the disjoint short cycles has only a bound of order r. So we cannot count the blocks with a short cycle without a packing bound.
- *A higher girth in [M, Theorem 7.1].* See fact 3 above.

7 Checks and files

All files are in `paper/minimal/`.

File	Content and result
`research/apply_one_scale.py`	Applies the first proof to a copy of the paper (Sections 8 and 1). Each replacement must match one time. The copy builds: 42 pages, 1615 lines, no undefined reference, no overfull line.
`research/apply_two_regions.py`	Applies the second proof to a copy. The copy builds: 42 pages, 1644 lines, no undefined reference, no overfull line.
`research/trial-one-scale.pdf` `research/trial-fixed-girth.pdf`	The two copies of the paper. Section 8 starts on page 34.
`checks/tjoin_check.py`	Lemma 4.1 (i) and the probabilities of Lemma 4.1 for one and two regions, on 300 random subcubic graphs with 8 to 13 vertices; the ladders with 3 to 10 rungs. Exit code 0 for the seeds 1 and 7.
`checks/tube_cyclic_cuts.py`	Lemma 5.1 for the Heawood graph and $M = 4, 5$: girth 6, every set of at most four edges tested, two links between the rings 0 and 2. Exit code 0.
`checks/necklace_parity.py`	The parity statement of Section 6 for $4 \le M \le 7$. Exit code 0.

The scripts test the statements on small graphs. They do not test the proofs of Sections 2 and 4,

whose constants are too large for a computer.

The choice. The first proof is the smaller change: [M, Lemma 8.2] stays, and the section is shorter. The second proof has the cleaner statements and the stronger lemma, and the section is longer. Each of them needs a decision by KNT before the manuscript changes, and each makes the text differ more from the formal proof. My recommendation is the first proof for the manuscript. Theorem 4.4 then stays in this note; it is the start for the open question of Section 6.

References

[M] KNT, Astra and Fable 5.2, *The maximum number of cycles in a graph of large rank*, draft of 12 November 2026, 42 pages, file `paper/minimal/exact-maximum/exact-maximum.pdf`.